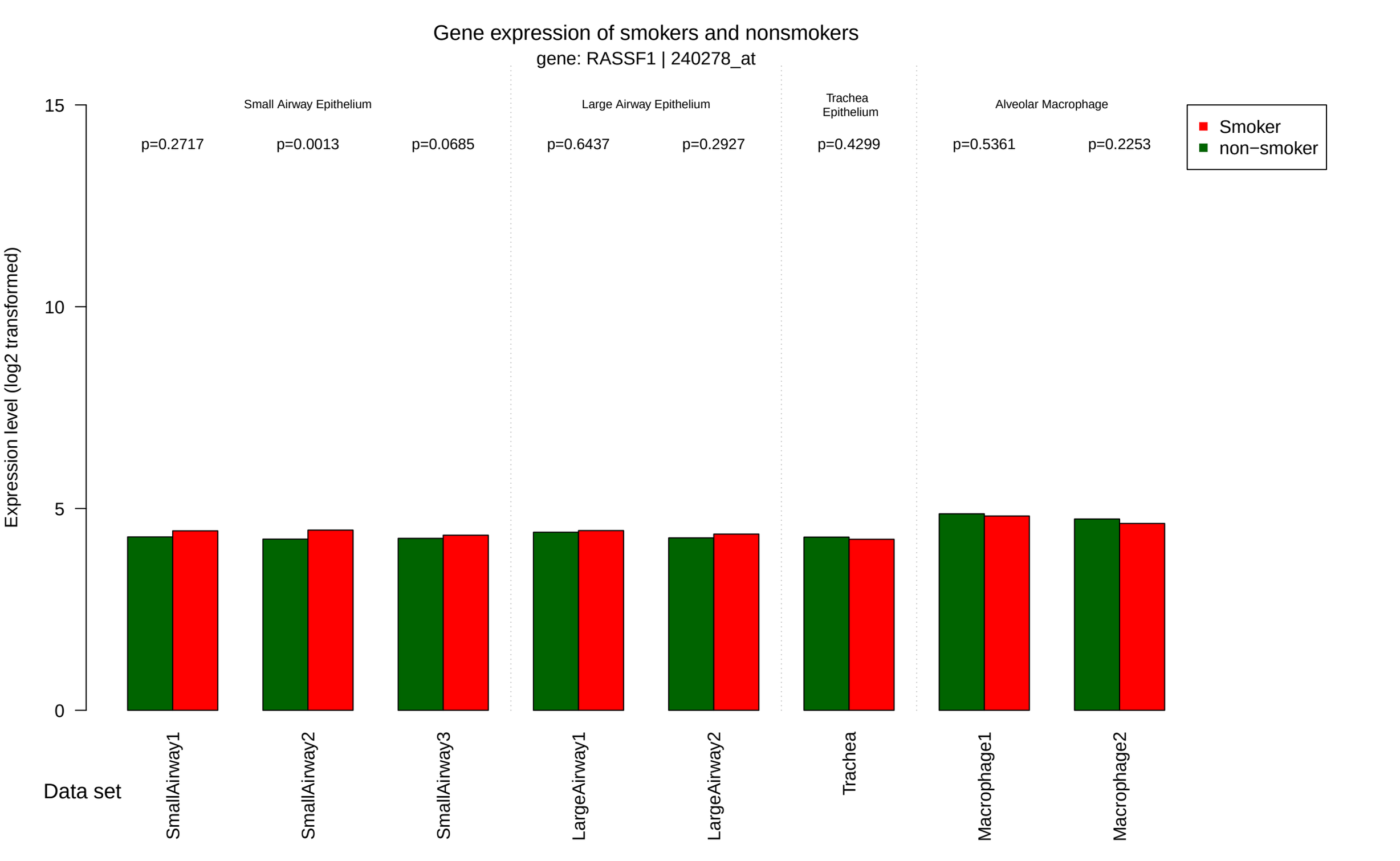
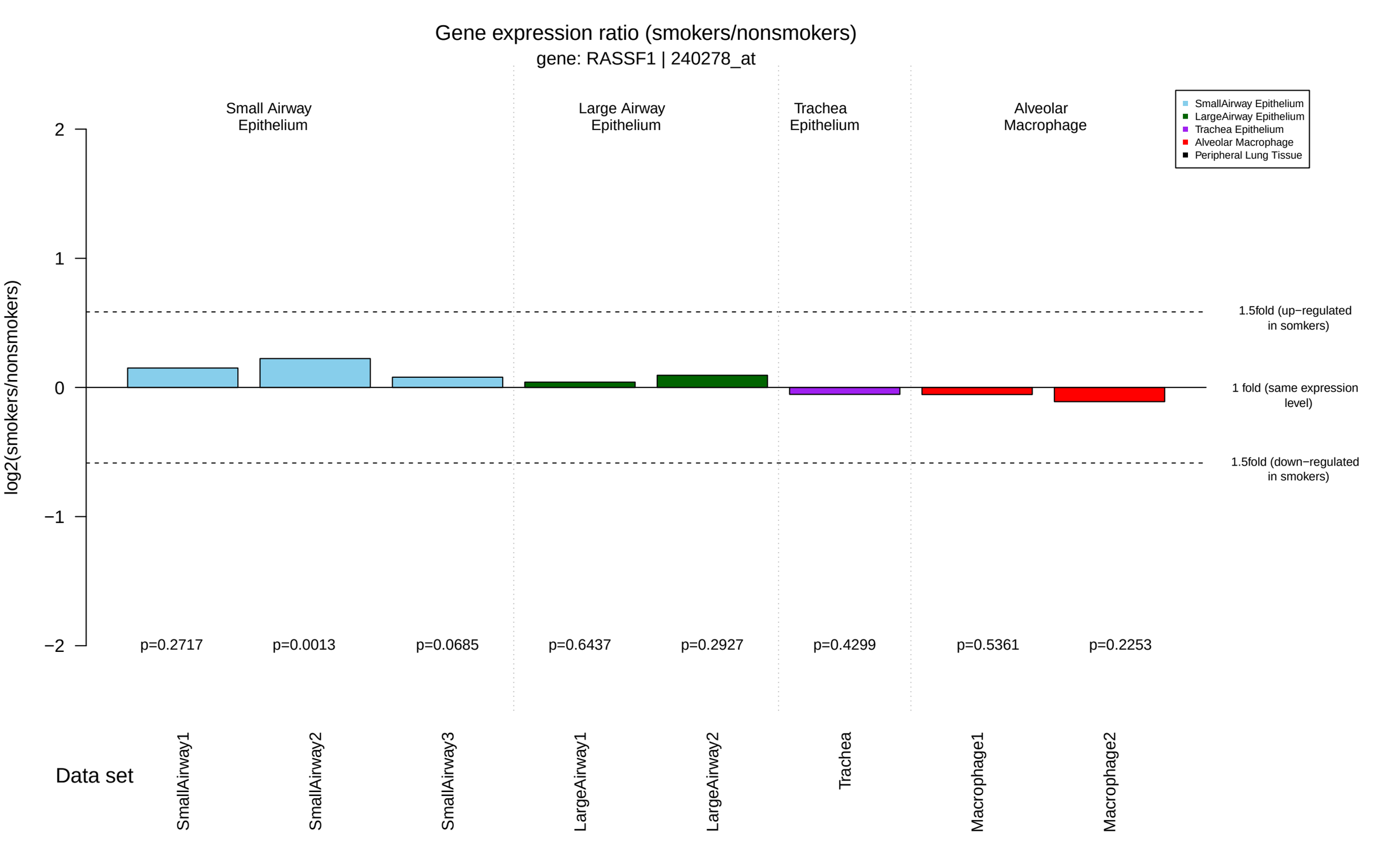
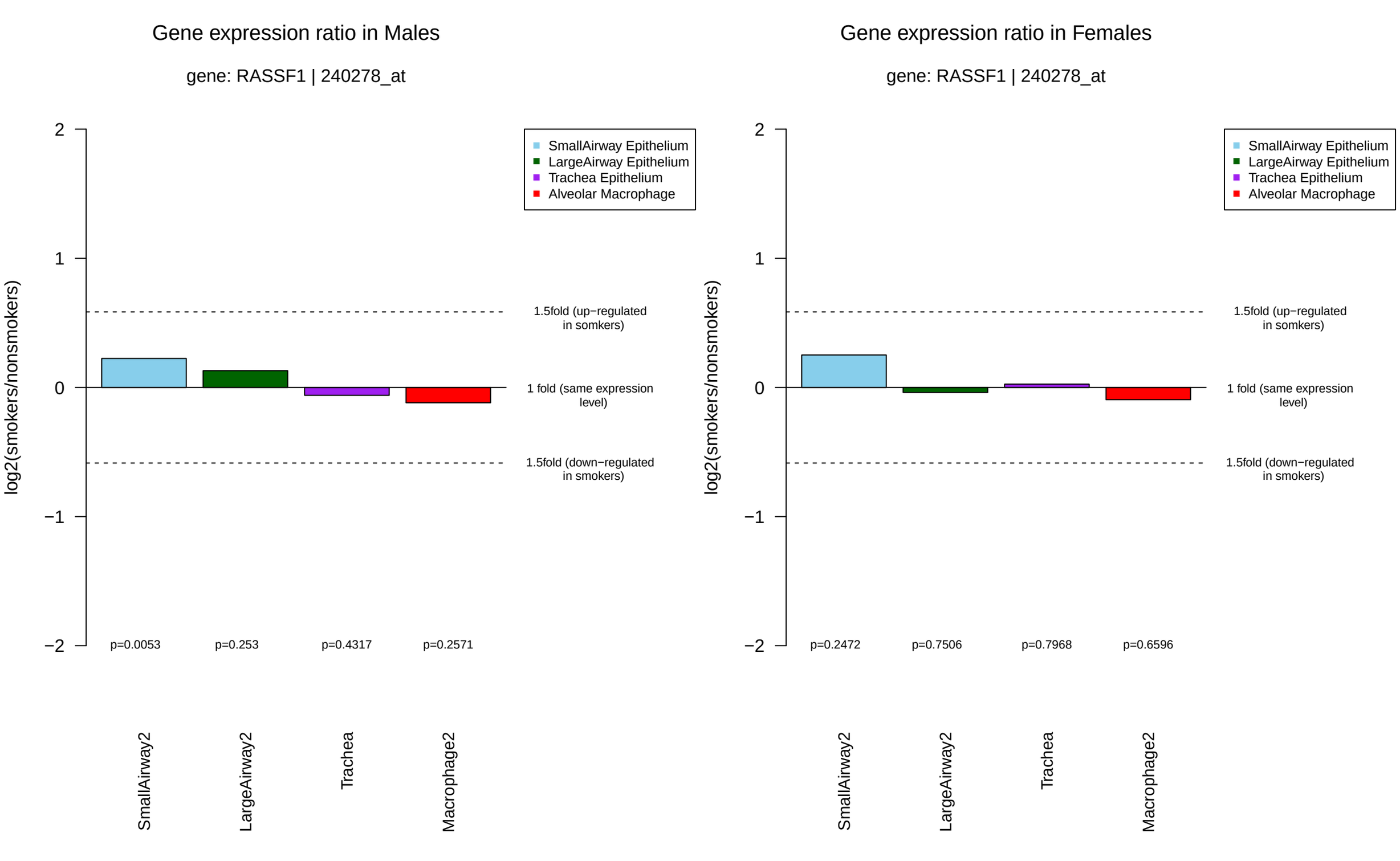
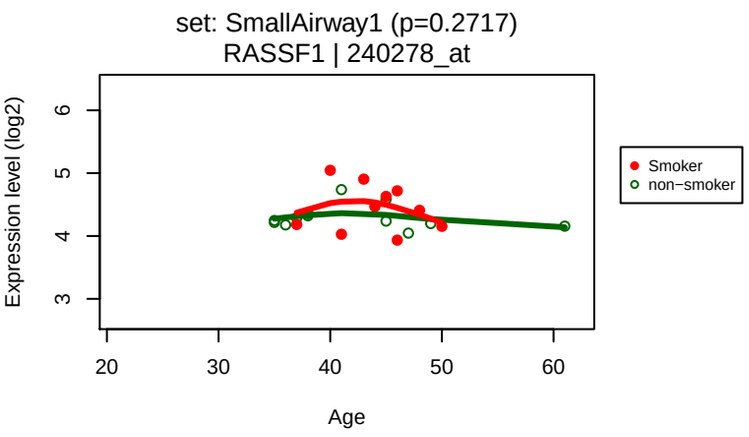
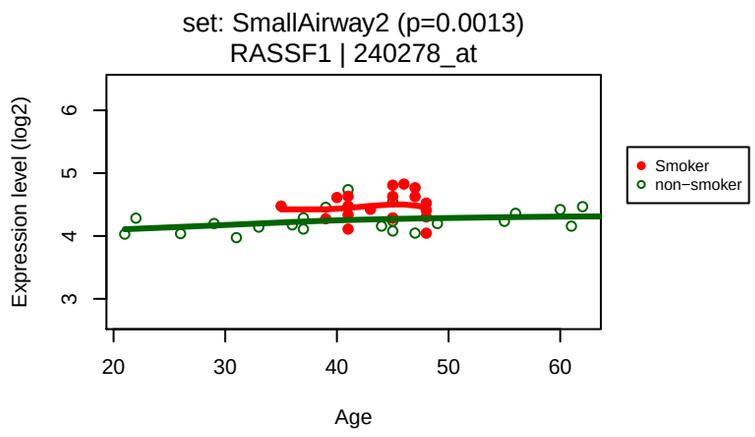
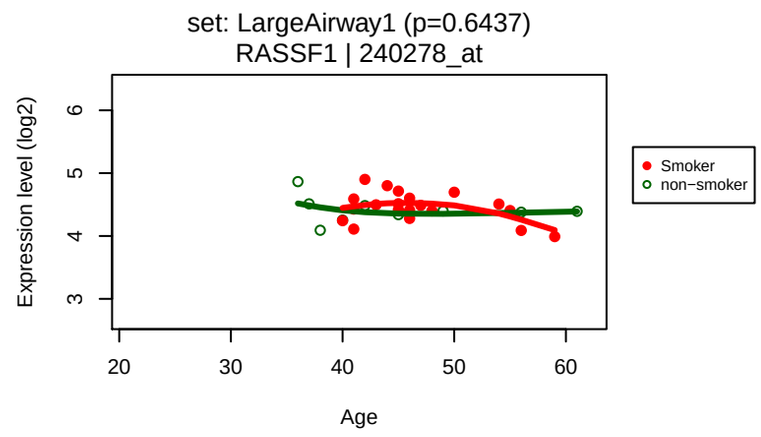

Gene expression of smokers and nonsmokers
gene: RASSF1 | 240278_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2717
p=0.0013
p=0.0685
p=0.6437
p=0.2927
p=0.4299
p=0.5361
p=0.2253
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

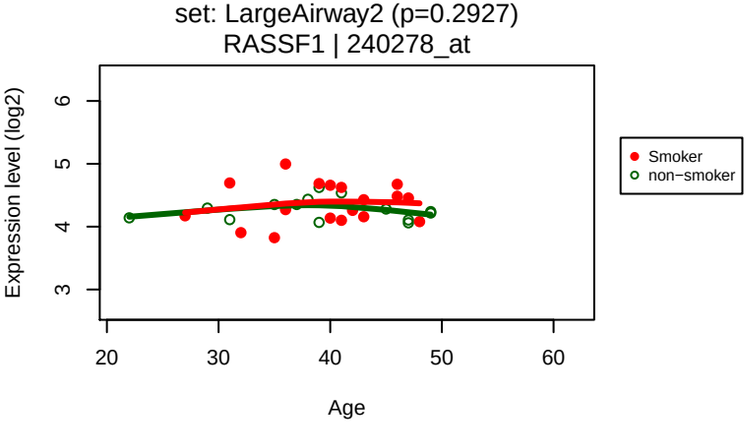

Gene expression ratio (smokers/nonsmokers)
gene: RASSF1 | 240278_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2717
p=0.0013
p=0.0685
p=0.6437
p=0.2927
p=0.4299
p=0.5361
p=0.2253
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

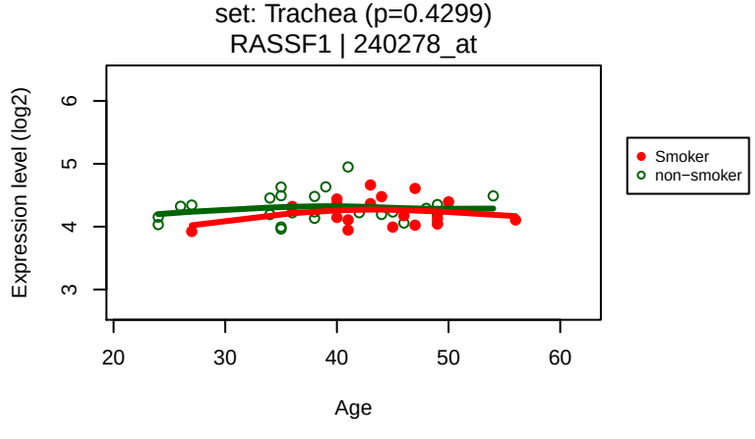

Gene expression ratio in Males
gene: RASSF1 | 240278_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0053
p=0.253
p=0.4317
p=0.2571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASSF1 | 240278_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2472
p=0.7506
p=0.7968
p=0.6596
SmallAirway2
LargeAirway2
Trachea
Macrophage2

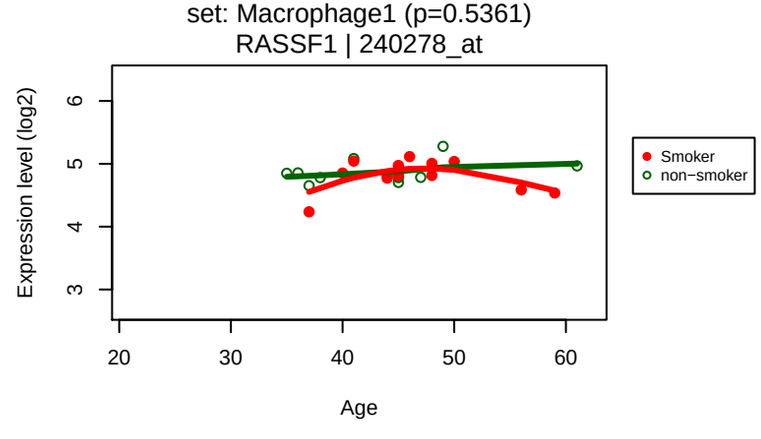

set: SmallAirway1 (p=0.2717)
RASSF1 | 240278_at
Expression level (log2)
Age
Smoker
non−smoker

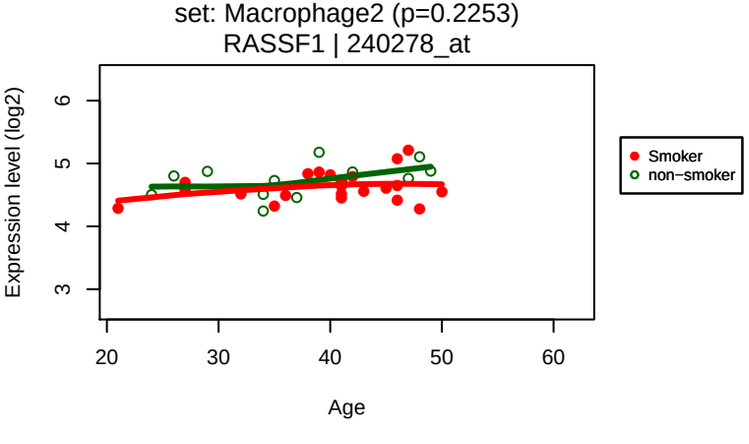

set: SmallAirway2 (p=0.0013)
RASSF1 | 240278_at
Expression level (log2)
Age
Smoker
non−smoker

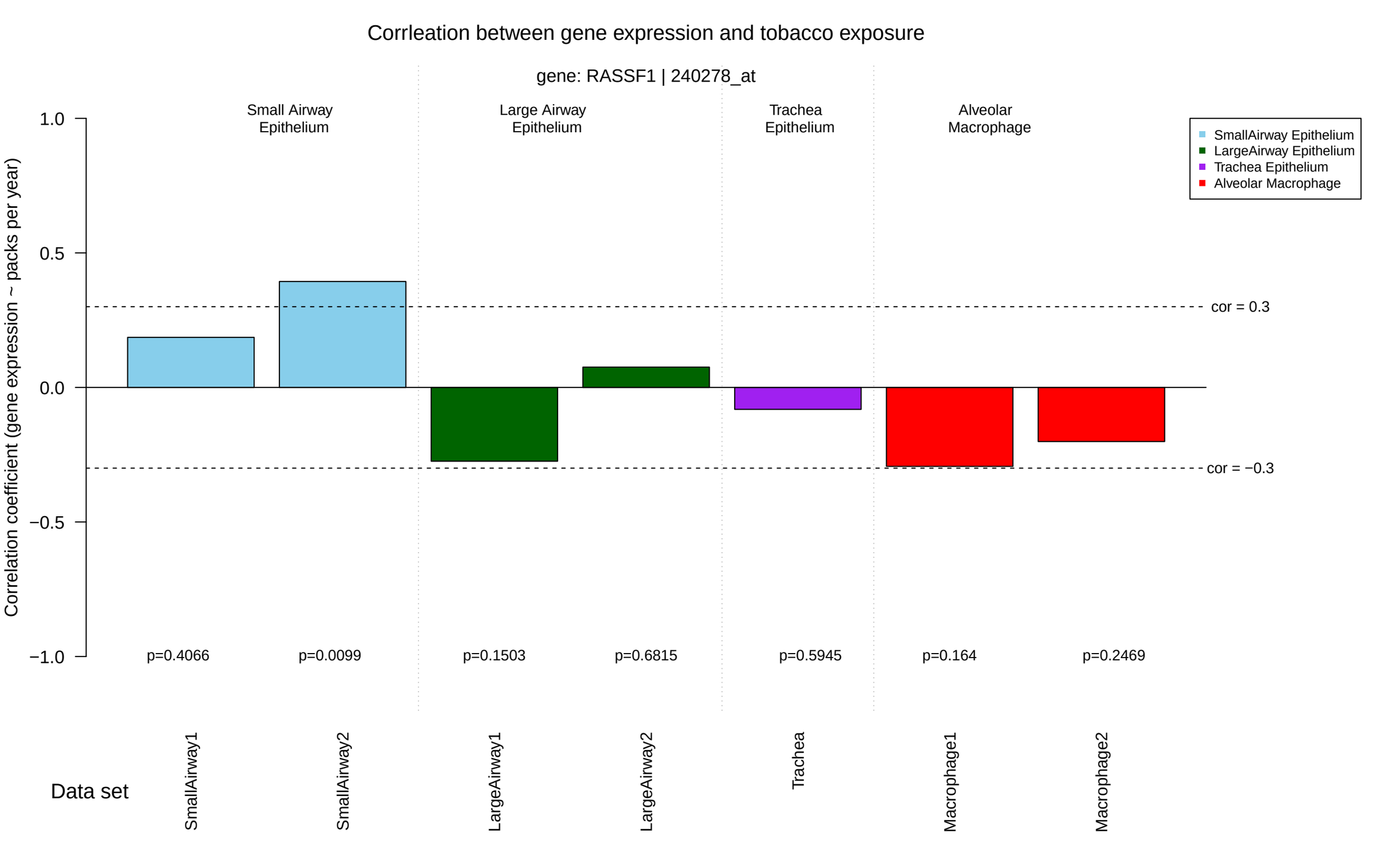

set: LargeAirway1 (p=0.6437)
RASSF1 | 240278_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2927)
RASSF1 | 240278_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4299)
RASSF1 | 240278_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5361)
RASSF1 | 240278_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2253)
RASSF1 | 240278_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASSF1 | 240278_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4066
p=0.0099
p=0.1503
p=0.6815
p=0.5945
p=0.164
p=0.2469
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2